

parametric upper limb and hand 3d print prosthetic product design via anthropometric data and artificial intelligence

Research shows that parametric design approaches combining anthropometric data (such as grip circumference and palm breadth) with AI methods can generate personalized 3D-printed upper limb prostheses with proven functional performance.

Abstract

Ten studies demonstrate that parametric techniques can tailor upper limb and hand prostheses using anthropometric data and, in some cases, artificial intelligence. Four studies detail the use of specific measurements—such as grip circumference, palm breadth, thumb dimensions, and wrist range of motion—to generate anatomically personalized designs via Computer-Aided Design protocols (e.g., Solidworks and Blender). Two studies employ AI or data-driven methods: one classifies electromyographic signals for adaptive control (yielding a 7.4/10 BAM-ULA score with 100% gross motor task success) and another applies multi-index optimization for socket design.

Manufacturing methods predominantly rely on three-dimensional printing techniques—fused deposition modeling, selective laser sintering, and desktop printing—with materials including PLA, ABS, and nylon. Reported costs range from low material expenditures (approximately US\$50) to complete devices under US\$350. Target populations span transradial amputees, children, and users in low-income or developing regions. These studies indicate that parametric design supported by anthropometric data—and, when applied, selective AI approaches—can produce adjustable, cost-effective prosthetic solutions with quantifiable functional outcomes.

Paper search

We performed a semantic search using the query “parametric upper limb and hand 3d print prosthetic product design via anthropometric data and artificial intelligence” across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We retrieved the 50 papers most relevant to the query.

Screening

We screened in sources that met these criteria:

- **Upper Limb Prosthetics Focus:** Does this study focus on upper limb prosthetics (including partial hand, wrist, forearm, or full arm prosthetics)?
- **3D Printing Technology:** Does this study involve 3D printing/additive manufacturing technology for prosthetic production?
- **Parametric Design Approach:** Does this study incorporate parametric design approaches in prosthetic development?
- **Anthropometric Data Utilization:** Does this study utilize anthropometric data in the prosthetic design process?
- **AI/ML Integration:** Does this study employ artificial intelligence, machine learning, or automated design algorithms in prosthetic development?
- **Human Validation:** Does this study involve human participants (amputees, prosthetic users) or include design validation with the target population?

- **Relevant Outcomes:** Does this study report on design methodology, functional outcomes, fit assessment, or user satisfaction?
- **Methodological Rigor:** Is this study a full research article with complete methodology (not an opinion piece, editorial, or conference abstract without full methodology)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Parametric Design Approach:**

Extract details about the parametric design methodology including:

- Which anthropometric parameters were used (e.g., palm breadth, grip circumference, hand length)
- How parameters were selected or validated (statistical analysis, biomechanical studies)
- Parameter relationships and dependencies (which parameters drive others)
- Number of design variables and their ranges
- Parametric modeling software or methods used
- Whether design scales automatically or requires manual adjustment

- **Anthropometric Data Collection:**

Extract methods for obtaining body measurements including:

- Data collection technique (3D scanning, photography, manual measurement, existing databases)
- Equipment used (smartphones, 3D scanners, measuring tools)
- Processing algorithms (Statistical Shape Models, computer vision, manual digitization)
- Accuracy/error rates of measurement methods
- Sample size and demographics of anthropometric data
- Which body segments or features were measured

- **AI Implementation:**

Extract details about artificial intelligence integration including:

- AI application area (design optimization, control systems, customization, measurement processing)
- Machine learning algorithms used (classification, regression, neural networks)
- Input data types (EMG signals, images, sensor data)
- Training data and methods
- AI performance metrics (accuracy, response time, success rates)
- Whether AI learns/adapts to individual users

- **3D Printing Specifications:**

Extract 3D printing technical details including:

- Printing technology (FDM, SLA, SLS, material extrusion)
- Materials used and their properties
- Print settings and parameters
- Assembly requirements (single print vs multiple parts)

- Post-processing steps required
- Print time and material costs
- Printing challenges and solutions identified

- **Functional Performance:**

Extract quantitative performance results including:

- Grip/pinch force capabilities (N or kg)
- Range of motion measurements
- Control accuracy and response time
- Battery life or power consumption
- Standardized assessment scores (BAM-ULA, DASH, etc.)
- Comparison to commercial prosthetics or benchmark values
- Durability testing results

- **User Testing Results:**

Extract user evaluation outcomes including:

- Number and demographics of test participants
- Testing protocol or tasks performed
- User satisfaction ratings or feedback
- Ease of use and learning curve
- Daily living activity performance
- User preferences for design features
- Acceptance rates and reasons for rejection/adoption

- **Design Principles:**

Extract key design considerations including:

- Target user population (age, amputation level, geographic region)
- Cost targets and actual costs achieved
- Manufacturing simplicity requirements
- Modularity and customization capabilities
- Accessibility features for low-resource settings
- Bio-inspired design elements incorporated
- Trade-offs between functionality and simplicity

Results

Characteristics of Included Studies

Study	Study Focus	Design Approach	Manufacturing Method	Target Population	Full text retrieved
Romero et al., 2025	Artificial intelligence-driven, three-dimensional printed myoelectric prosthesis with anatomical personalization	Parametric modeling, artificial intelligence-based control	Additive manufacturing (three-dimensional printing), three-dimensional scanning	Transradial amputees, low-income regions	No
Lopez, 2021	Accessible, three-dimensional printed hand prostheses for low-income settings	Parametric design, non-assembly fabrication, bio-inspired	Material extrusion (fused deposition modeling), smartphone-based three-dimensional modeling	Transradial amputees, low-income/developing countries	No
Moreo, 2016	Parametric, body-powered three-dimensional printed hand for children	Parametric design, statistical analysis of anthropometry	Three-dimensional printing (no mention found), modular assembly	Children in developing countries	No
Saldarriaga et al., 2024	Parametric design for affordable, customized three-dimensional printed sockets	Parametric design, customization	Three-dimensional printing with polylactic acid (PLA)	Transradial congenital amputees	No
Fajardo et al., 2020	Galileo Hand: modular, parametric, three-dimensional printed myoelectric prosthesis	Modular, parametric, self-contained, user feedback	Three-dimensional printing (nylon, acrylonitrile butadiene styrene (ABS), polylactic acid (PLA)), multi-part assembly	Transradial amputees, developing countries	Yes

Study	Study Focus	Design Approach	Manufacturing Method	Target Population	Full text retrieved
Nini et al., 2024	Parametric computer-aided design (CAD) for custom prosthetic fingers	Parametric CAD, anthropometric modeling	Selective laser sintering (SLS) three-dimensional printing, three-dimensional scanning	Adults (tested on one volunteer)	No
Zuniga et al., 2015	Cyborg Beast: low-cost, three-dimensional printed hand for children	Parametric scaling, distance fitting, open-source	Fused deposition modeling (PLA, ABS), multi-part assembly	Children with upper-limb reductions	Yes
Fajardo et al., 2017	Open-source, multifunctional, modular three-dimensional printed prosthesis	Modular, parametric, self-contained, surface electromyography (sEMG) control	Three-dimensional printing (no mention found)	Transradial amputees (various levels)	No
Cabibihan et al., 2018	Patient-specific prosthetic arms via computed tomography (CT), computer-aided design (CAD), three-dimensional printing	Mirrored geometry, high-accuracy fit	Desktop three-dimensional printing, silicone casting	Upper limb amputees (no further specification)	No
Gu et al., 2024	Data-driven, multi-index optimization for socket design	Data-driven, multi-index optimization, clinical criteria	No mention found	Forearm amputees	No

Design Approaches

- Parametric design/modeling/scaling : Found in 8 studies.
- Modularity : Used in 2 studies.
- Artificial intelligence or data-driven approaches : Used in 2 studies.
- Anthropometric or statistical modeling : Used in 2 studies.
- Customization : Focus in 2 studies.
- Other approaches : Open-source, user feedback, surface electromyography/artificial intelligence control, non-assembly, and bio-inspired approaches each found in 1 study.

Manufacturing Methods

- Three-dimensional printing (general or no mention found) : Used in 6 studies.
- Fused deposition modeling/material extrusion : Used in 2 studies.
- Selective laser sintering : Used in 1 study.
- Three-dimensional scanning : Used in 2 studies.
- Modular or multi-part assembly : Used in 2 studies.
- Materials : Polylactic acid (PLA) specified in 3 studies, acrylonitrile butadiene styrene (ABS) in 2, nylon in 1.
- Other methods : Smartphone-based and desktop three-dimensional printing each used in 1 study; silicone casting in 1 study.
- No mention found : Manufacturing method not described in 1 study.

Target Populations

- Transradial amputees (including forearm) : Target in 6 studies.
- Children : Target in 2 studies.
- Low-income or developing regions/countries : Focus in 4 studies.
- Congenital amputees : Target in 1 study.
- Adults : Target in 1 study (tested on a single volunteer).
- Upper limb amputees (no further specification) : 1 study.
- All studies provided some population information.

Thematic Analysis

Parametric Design Methodologies

Study	Parametric Design Details
Romero et al., 2025	Uses parametric modeling for anatomical personalization; no mention found of specific parameters or software
Lopez, 2021	Parametric socket design from two-dimensional photos using statistical shape model (SSM); manual adjustment required due to measurement errors
Moreo, 2016	Four key anthropometric parameters (grip circumference, palm breadth, thumb breadth, palm length) selected via statistical analysis; all linked to palm breadth; Solidworks computer-aided design (CAD) with external file for easy adjustment
Saldarriaga et al., 2024	Parametric and transradial prosthesis design; no mention found of specific parameters or software
Fajardo et al., 2020	Modular, parametric design allowing modification of phalanges' length; validated via user feedback; no mention found of software

Study	Parametric Design Details
Nini et al., 2024	Parametric computer-aided design (CAD) for finger design based on anthropometric parameters from three-dimensional scans; no mention found of software
Zuniga et al., 2015	Parametric scaling using hand/forearm length, palm width, wrist range of motion from photos; validated against direct measurement; Blender 7.2 and sizing chart for scaling
Fajardo et al., 2017	Modular, parametric design; no mention found of specific parameters or software
Cabibihan et al., 2018	Mirrored geometry from computed tomography (CT) scans; no mention found of software; focus on high-accuracy fit
Gu et al., 2024	Data-driven, multi-index optimization; no mention found of specific parameters or software

Anthropometric Parameters:

- Four studies (Moreo, Fajardo 2020, Nini, Zuniga) detailed specific anthropometric parameters, most commonly grip circumference, palm breadth, thumb breadth, palm length, hand/forearm length, palm width, and wrist range of motion.
- Five studies (Romero, Lopez, Saldarriaga, Fajardo 2017, Gu) did not mention specific parameters.

Software:

- Only two studies specified the software used for parametric design (Moreo: Solidworks; Zuniga: Blender). The other eight studies did not mention software.

Input Methods:

- Two studies used two-dimensional photos (Lopez, Zuniga), one used three-dimensional scans (Nini), and one used computed tomography (CT) scans (Cabibihan).

Statistical or Data-Driven Approaches:

- Two studies (Lopez, Gu) used statistical or data-driven approaches, including statistical shape modeling.

Modular or Adjustable Designs:

- Three studies (Moreo, Fajardo 2020, Fajardo 2017) described modular or easily adjustable designs.

Validation:

- Two studies (Fajardo 2020, Zuniga) reported validation of the parametric design via user feedback or direct measurement.

Other Notable Approaches:

- One study (Cabibihan) focused on high-accuracy fit.
- One study (Romero) mentioned anatomical personalization.

Anthropometric Data Integration and Customization

Study	Data Collection & Processing	Accuracy/Validation	Customization Approach
Romero et al., 2025	Three-dimensional scanning; parametric modeling	No mention found	Anatomical personalization
Lopez, 2021	Smartphone photos, statistical shape model (SSM), manual measurement	Automatic method error too large; manual method used	Parametric socket design
Moreo, 2016	No mention found; parameters identified	No mention found	Parametric design for children
Saldarriaga et al., 2024	Anthropometric measurements (no mention found of method)	No mention found	Custom socket for one user
Fajardo et al., 2020	User feedback from volunteers	No mention found	Modular, customizable design
Nini et al., 2024	Three-dimensional scanning of phalanges	No mention found	Custom finger design
Zuniga et al., 2015	Direct measurement, photography, ImageJ	No significant difference between methods; validated	Parametric scaling, distance fitting
Fajardo et al., 2017	No mention found	No mention found	Modular, parametric design
Cabibihan et al., 2018	Computed tomography (CT) imaging, computer-aided design (CAD)	High accuracy (Dice similarity coefficient 0.95–0.96, less than 1% error)	Mirrored geometry for fit
Gu et al., 2024	No mention found; motion/pressure data	No mention found	Data-driven socket optimization

Data Collection & Processing:

- Three-dimensional scanning: 2 studies.
- Imaging (computed tomography): 1 study.
- Photography (including smartphone photos): 2 studies.
- Direct or manual measurement (including anthropometric): 3 studies.
- User feedback: 1 study.
- Statistical shape modeling: 1 study.
- Motion/pressure data: 1 study.
- No mention found of data collection or processing method in 3 studies.

Accuracy/Validation:

- High accuracy or validation: 2 studies.

- Error too large for automatic methods, manual methods used: 1 study.
- No mention found of accuracy or validation in 7 studies.

Customization Approach:

- Parametric design (including modeling, scaling, or socket design): 4 studies.
- Anatomical personalization (including mirrored geometry): 2 studies.
- Modular or customizable design: 2 studies.
- Data-driven optimization: 1 study.
- Customization for one user (including custom finger design): 2 studies.

Three-Dimensional Printing Technologies and Manufacturing Constraints

Study	Printing Technology	Materials	Assembly/Post-processing	Cost/Accessibility
Romero et al., 2025	No mention found	No mention found	No mention found	Affordable, accessible
Lopez, 2021	Material extrusion (fused deposition modeling)	Three-dimensional printed polymers	Single print, minimal post-assembly	Low-cost, single printer
Moreo, 2016	No mention found	No mention found	3 parts, glued/elastic bands	Price requirement met
Saldarriaga et al., 2024	No mention found (likely fused deposition modeling)	Polylactic acid (PLA)	No mention found	Affordable, optimized design
Fajardo et al., 2020	No mention found	Nylon, acrylonitrile butadiene styrene (ABS), polylactic acid (PLA)	Multi-part, elastics/strings	Less than \$350, accessible
Nini et al., 2024	Selective laser sintering (SLS)	No mention found	No mention found	Cost reduction via additive manufacturing
Zuniga et al., 2015	Fused deposition modeling (Makerbot, Ultimaker)	Polylactic acid (PLA), acrylonitrile butadiene styrene (ABS)	Multi-part, 2.5 hours assembly	\$50 materials, online files
Fajardo et al., 2017	No mention found	No mention found	No mention found	Cost/time reduction
Cabibihan et al., 2018	Desktop three-dimensional printing	No mention found (molds), silicone casting	Molds, casting	No mention found
Gu et al., 2024	No mention found	No mention found	No mention found	No mention found

Printing Technology:

- Fused deposition modeling or material extrusion: 2 studies.
- Selective laser sintering: 1 study.
- Desktop three-dimensional printing: 1 study.
- No mention found of printing technology in 6 studies (including 1 described as "likely fused deposition modeling").

Materials:

- Polylactic acid (PLA): 3 studies.
- Acrylonitrile butadiene styrene (ABS): 2 studies.
- Nylon: 1 study.
- Three-dimensional printed polymers: 1 study.
- Silicone (via casting): 1 study.
- No mention found of materials in 6 studies.

Assembly/Post-processing:

- Single print or minimal post-assembly: 1 study.
- Multi-part assembly (including use of elastics, strings, or glue): 3 studies.
- Molds and casting: 1 study.
- No mention found of assembly or post-processing in 5 studies.

Cost/Accessibility:

- Statements of affordability, low cost, price requirement met, cost reduction, or optimized design: 7 studies.
- Specific price reported: 2 studies ("less than \$350" and "\$50 materials").
- Accessibility features (such as online files or use of a single printer): 3 studies.
- No mention found of cost or accessibility in 2 studies.

Data-Driven Optimization and Artificial Intelligence Applications

Study	Artificial Intelligence Application Area	Algorithms/Methods	Input Data	Adaptation/Personalization
Romero et al., 2025	Control system (electromyography classification)	Classification	Electromyography signals	Adapts to individual users
Lopez, 2021	Customization (socket modeling)	Statistical shape model (SSM), algorithm	Two-dimensional photos	No adaptation beyond initial fit
Moreo, 2016	Design optimization	Statistical analysis	No mention found	Parametric personalization
Saldarriaga et al., 2024	No mention found	No mention found	No mention found	No mention found
Fajardo et al., 2020	Control system (electromyography)	Threshold, finite state machine	Electromyography signals	No mention found
Nini et al., 2024	No mention found	No mention found	No mention found	No mention found

Study	Artificial Intelligence Application Area	Algorithms/Methods	Input Data	Adaptation/Personalization
Zuniga et al., 2015	Measurement processing	No mention found	Photographs	No mention found
Fajardo et al., 2017	Control system (surface electromyography)	No mention found	Electromyography signals	No mention found
Cabibihan et al., 2018	Measurement processing	No mention found	Computed tomography images	No mention found
Gu et al., 2024	Design optimization	Data-driven, multi-index optimization	Motion/pressure data, clinical criteria	No mention found

Artificial Intelligence Application Area:

- Three studies focused on control systems using electromyography or surface electromyography signals.
- Two studies addressed design optimization.
- Two studies focused on measurement processing.
- One study focused on customization (socket modeling).
- No mention found of application area for two studies.

Algorithms/Methods and Input Data:

- One study used classification, one used statistical shape modeling, one used statistical analysis, one used thresholding, one used finite state machine, and one used data-driven multi-index optimization. Five studies did not mention the algorithm or method.
- Electromyography signals were used as input data in three studies.
- Other input data included two-dimensional photos (one study), photographs (one study), computed tomography images (one study), motion/pressure data (one study), and clinical criteria (one study). No mention found of input data for three studies.

Adaptation/Personalization:

- One study adapted to individual users.
- One study used parametric personalization.
- One study reported no adaptation beyond initial fit.
- No mention found of adaptation or personalization for seven studies.

Performance Validation and Functional Outcomes

Study	Performance Metrics	Validation Method	Key Findings
Romero et al., 2025	Bilateral Assessment Measure for Upper Limb Amputees (BAM-ULA) score (7.4/10), gross/fine motor task success	BAM-ULA protocol, 9 users	Comparable to high-end devices; 100% gross, 22.2% fine motor success
Lopez, 2021	No mention found	Mechanical/user testing	Bio-inspired hand achieved superior pinch force; no quantitative data
Moreo, 2016	Pinch force (35 newtons), daily activity	Mechanical/user testing	Exceeds required force; functional for daily activities; durability needs improvement
Saldarriaga et al., 2024	No mention found	Preliminary evaluation, 1 user	Custom socket fabricated; no quantitative results
Fajardo et al., 2020	Grip force (2.5–5 kilograms/finger, 50 newtons power), range of motion, controller response	Mechanical/user testing, National Aeronautics and Space Administration Task Load Index (NASA TLX), 10 users	High accuracy, responsive control, user-friendly, less than \$350
Nini et al., 2024	Finite element method (FEM) analysis, trajectory comparison	Technical verification, 1 user	Mechanism robust; trajectory matches human finger
Zuniga et al., 2015	Range of motion, user survey	Direct/photo measurement, 11 children	Positive impact on quality of life, daily use; validated measurement
Fajardo et al., 2017	No mention found	Mechanical/control testing	High accuracy/responsiveness; no quantitative data
Cabibihan et al., 2018	Dice similarity coefficient, percent error, mean distance	Shape/fit validation, 1 user	High accuracy (Dice similarity coefficient 0.95–0.96, less than 1% error); user satisfied
Gu et al., 2024	Muscle fatigue, clinical scale scores	5 users, load conditions	Reduced fatigue, modest score improvement

Performance Metrics:

- Force or strength measurements: 3 studies.
- Range of motion: 2 studies.
- Accuracy or fit metrics: 1 study.
- Task success (gross/fine motor): 1 study.
- User-reported outcomes (such as surveys, fatigue, or clinical scales): 2 studies.
- Technical or engineering metrics (such as finite element method analysis or trajectory comparison): 1 study.
- No mention found of performance metrics in 3 studies.

Validation Methods:

- Mechanical testing: 4 studies.
- User testing: 9 studies; of these, 3 studies used only 1 user, and the others used 5, 9, 10, or 11 users.
- Technical or engineering validation: 2 studies.
- Direct or photo measurement: 1 study.
- Control testing: 1 study.
- National Aeronautics and Space Administration Task Load Index (NASA TLX): 1 study.

Key Findings:

- Functional success (such as gross/fine motor task completion or daily activity): 4 studies.
- Exceeding required force or strength: 2 studies.
- High accuracy or fit: 3 studies.
- User satisfaction or positive impact: 4 studies.
- Cost information: 1 study (device costing less than \$350).
- No quantitative results: 3 studies.
- Durability limitation: 1 study.
- Technical success (such as robust mechanism or human-like trajectory): 1 study.
- Comparison to high-end devices: 1 study.

References

- Alejandro M. Saldarriaga, Enzo Romero, Victoria E. Abarca, and Dante A. Elías. “A Parametric Design Approach for Affordable Customized 3D Socket for Transradial Upper Limb Prostheses.” *International Conference on Control, Decision and Information Technologies*, 2024.
- Enzo Romero, Jose G. Garcia, Magno Parra, Sebastian Caballa, Alejandro M. Saldarriaga, Edson F. Luque, Dante J. Rodriguez, Victoria E. Abarca, and Dante A. Elías. “An Affordable AI-Driven and 3D-Printed Personalized Myoelectric Prosthesis: Design, Development, and Assessment.” *IEEE Access*, 2025.
- J. Cabibihan, M. K. Abubasha, and N. Thakor. “A Method for 3-D Printing Patient-Specific Prosthetic Arms With High Accuracy Shape and Size.” *IEEE Access*, 2018.
- J. Lopez. “Accessible Hand Prostheses: 3D Printing Meet Smartphones,” 2021.
- J. Zuniga, D. Katsavelis, Jean M Peck, John Stollberg, Marcin Petrykowski, Adam Carson, and Cristina Fernandez. “Cyborg Beast: A Low-Cost 3d-Printed Prosthetic Hand for Children with Upper-Limb Differences.” *BMC Research Notes*, 2015.
- Julio Fajardo, Victor Ferman, Ali Lemus, and E. Rohmer. “An Affordable Open-Source Multifunctional Upper-Limb Prosthesis with Intrinsic Actuation.” *Workshop on Advanced Robotics and Its Social Impacts*, 2017.
- Julio Fajardo, Victor Ferman, Diego Cardona, Guillermo Maldonado, Ali Lemus, and E. Rohmer. “Galileo Hand: An Anthropomorphic and Affordable Upper-Limb Prosthesis.” *IEEE Access*, 2020.
- Ludovica Nini, Alessandro Ceccarelli, N. Tagliamonte, Loredana Zollo, and Fabrizio Taffoni. “Parametric 3D Modeling of a Customized Prosthetic Hand Finger for Additive Manufacturing.” *International Conference on Biomedical Robotics and Biomechatronics*, 2024.
- M. Moreo. “Parametric Design of a 3D Printable Hand Prosthesis for Children in Developing Countries,” 2016.
- Yu Gu, Long He, Haozhou Zeng, Jiaxing Li, Ning Zhang, Xiufeng Zhang, and Tao Liu. “A Data-Driven Design Framework for Structural Optimization to Enhance Wearing Adaptability of Prosthetic Hands.” *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2024.